Time & Work



CONTENT

1. INTRODUCTION
2. BASIC CONCEPTS
3. RULES AND TRICKS
4. PRACTICE QUESTIONS

TIME & WORK

Time and Work deals with the time taken by an individual or a group of individuals to complete a piece of work and the efficiency of the work done by each of them.

The number of units of work done per unit time is called the **Rate of Work (R) OR Efficiency.**

Thus,

Work Done(W) = Time (T) * Rate of Work (R)

Total Work Done = Number of Days × Efficiency

BASIC CONCEPTS

While solving the problems based on TIME and WORK , we need to calculate :

1. **TIME :** Time taken in completing the assigned job.
2. **INDIVIDUAL'S TIME :** Time taken to complete the work alone.
3. **WORK DONE :** It actually means the total work that has to be done by an Individual or Group of people.

BASIC CONCEPT

Efficiency and Time are inversely proportional to each other.

Example: X : Y is the ratio of the number of men which are required to complete a piece of work, then the ratio of the time taken by them to complete the work will be Y :X.

QUESTIONS TO BE DISCUSSED

- To calculate the time when working together
- To calculate the time when working alone
- To calculate the fraction of work left
- Efficiency based problems
- Alternate assistance
- Person leaves / joins the work in between.

PROBLEM STATEMENT 1

A does a work in 10 days and B does the same work in 15 days. In how many days they together do the same work.

a) 5 days
b) 6 days
c) 8 days
d) 9 days



EXPLANATION

A does a work in 10 days and B does the same work in 15 days. In how many days they together do the same work.

a) 5 days
b) 6 days
c) 8 days
d) 9 days

SOLUTION

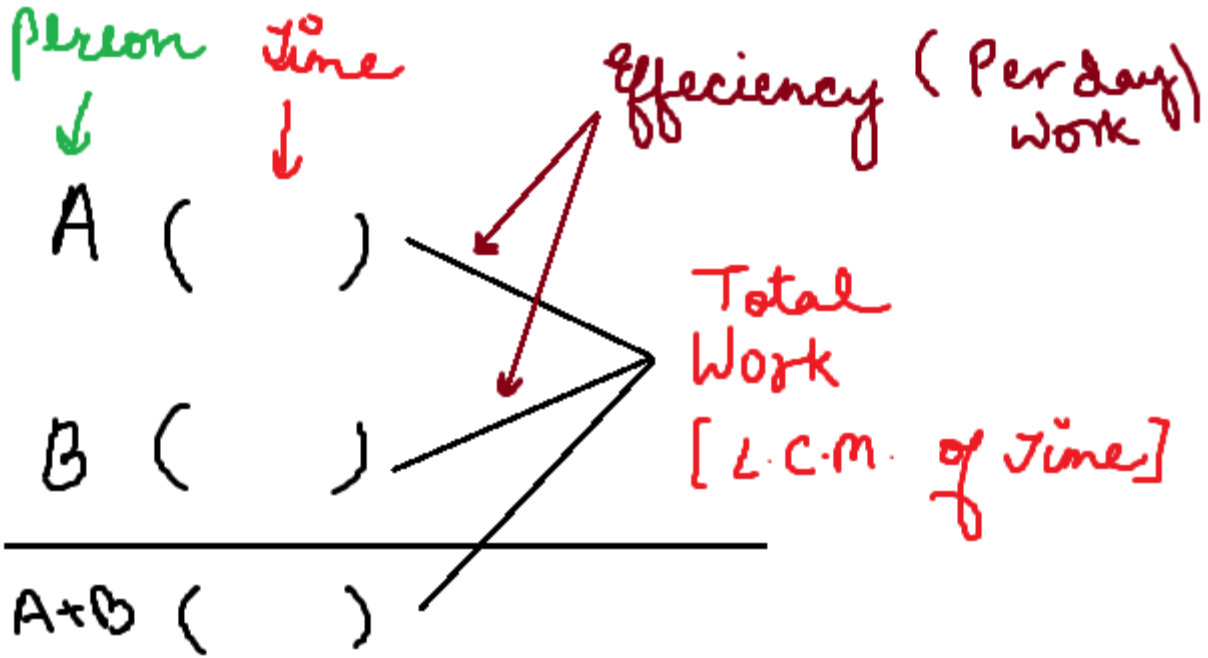
A can complete the work alone in 10 days
A's per day work = 1 day's work = 1/10

B can complete the work alone in 15 days.
B's per day work = 1 day's work = 1/15

Both Together (A+B)'s 1 day's work = $\left(\frac{1}{10} + \frac{1}{15}\right) = \frac{5}{30}$

Both together can com plete the work in = 6 days

RULES / TRICKS (LCM METHOD)

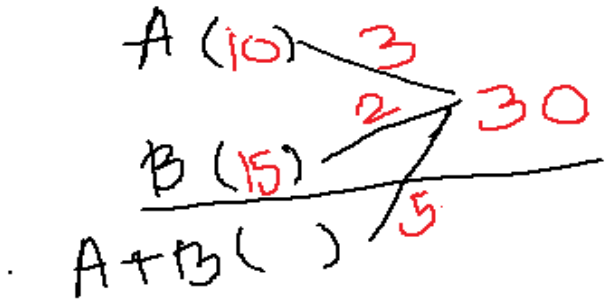


EXPLANATION

A does a work in 10 days and B does the same work in 15 days. In how many days they together do the same work.

a) 5 days
b) 6 days
c) 8 days
d) 9 days

SOLUTION



Total Time taken to complete the work together = $\frac{30}{5} = 6$ days

PROBLEM STATEMENT 2

A can do a piece of work in four days and B can do it in 12 days. In how many days will they finish the work, both working together?

a) 4 days
b) 6 days
c) 2 days
d) 3 days

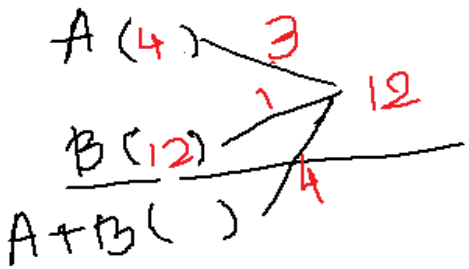


EXPLANATION

A can do a piece of work in four days and B can do it in 12 days. In how many days will they finish the work, both working together?

a) 4 days
b) 6 days
c) 2 days
d) 3 days

SOLUTION



Total Time taken to complete the work together = $\frac{12}{4} = 3$ days

PROBLEM STATEMENT 3

A takes 16 days to finish a job alone, while B takes 8 days to finish the same job. What is the ratio of their efficiency and who is less efficient.

a) 2:3, A

b) 2:3, B

c) 1:2, A

d) 1:2, B



EXPLANATION

Q3. A takes 16 days to finish a job alone, while B takes 8 days to finish the same job. What is the ratio of their efficiency and who is less efficient.

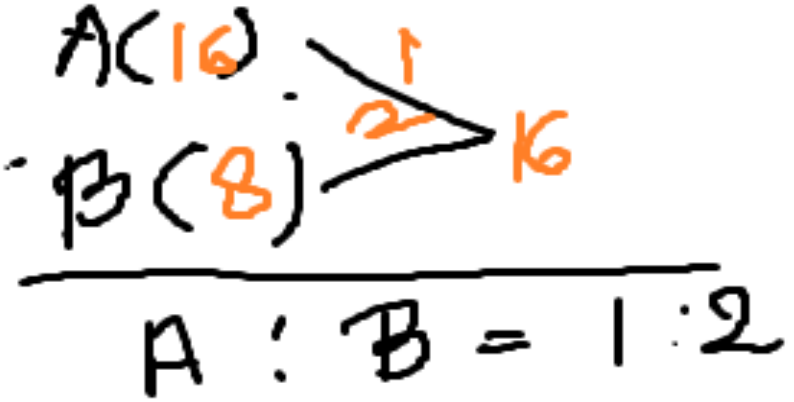
a) 2:3, A

b) 2:3, B

c) 1:2, A

d) 1:2, B

SOLUTION



B is more effecient than A.

PROBLEM STATEMENT 4

A, B and C can complete a piece of work in 24, 6 and 12 days respectively. Working together they can complete the same work in:

a) 1/24 day
b) 7/24 day
c) 24/7 days
d) 4 days



PROBLEM STATEMENT 5

A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is :

a) ¼
b) 1/10
c) 7/15
d) 8/15

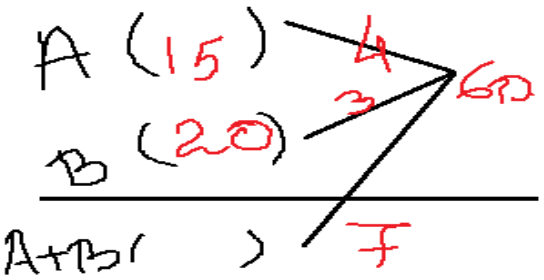


EXPLANATION

A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is :

a) ¼
b) 1/10
c) 7/15
d) 8/15

SOLUTION



(A+B) ⟶ 1 Day work ⟶ 7 units
(A+B) ⟶ 4 Days work ⟶ 7*4 units
= 28 units

Work Left = 60 - 28 = 32 units

Fraction of work left ⟶ $\frac{32}{60} = \frac{8}{15}$

PROBLEM STATEMENT 6

A can finish a work in 18 days and B can do the same work in half the time taken by A. Then working together, what Part of the same work they can finish in a day?

a) 1/6 days
b) 1/9 days
c) 2/5 days
d) 2/7 days

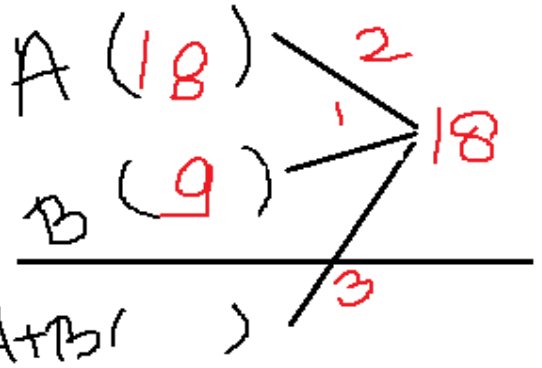


EXPLANATION

A can finish a work in 18 days and B can do the same work in half the time taken by A. Then working together, what Part of the same work they can finish in a day?

a) 1/6 days
b) 1/9 days
c) 2/5 days
d) 2/7 days

SOLUTION



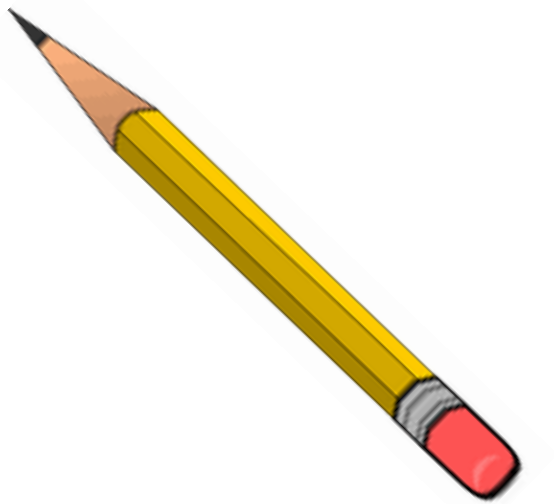
(A+B) ⟶ 1 Day work ⟶ 3 units

Fraction of work in 1 day ⟶ $\frac{3}{18} = \frac{1}{6}$

PROBLEM STATEMENT 7

P can complete a work in 12 days working 8 hours a day. Q can complete the same work in 8 days working 10 hours a day. If both P and Q work together, working 8 hours a day, in how many days can they complete the work?

a) 60/11
b) 61/11
c) 71/11
d) 72/11

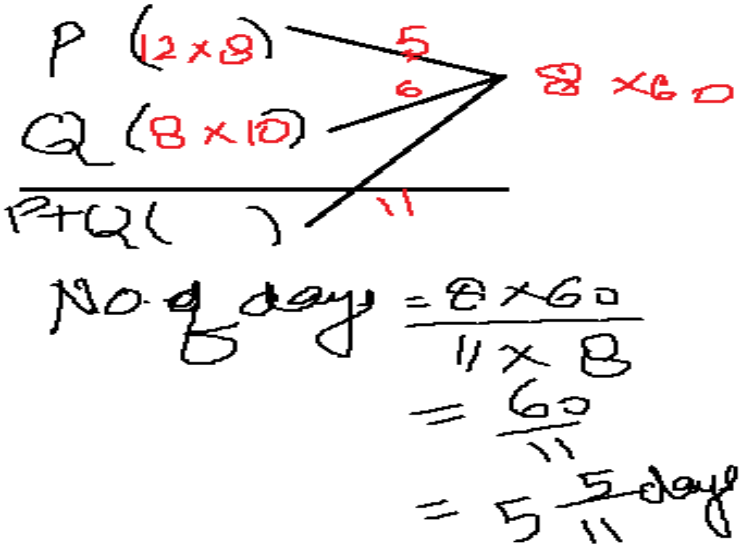


EXPLANATION

P can complete a work in 12 days working 8 hours a day. Q can complete the same work in 8 days working 10 hours a day. If both P and Q work together, working 8 hours a day, in how many days can they complete the work?

a) 60/11
b) 61/11
c) 71/11
d) 72/11

SOLUTION



PROBLEM STATEMENT 8

A and B can do a work in 12 days, B and C can do the same work in 15 days. C and A in 20 days. If A, B and C work together, they will complete the work in:

a) 5 days
b) 47 /6 days
c) 10 days
d) 47/3 days

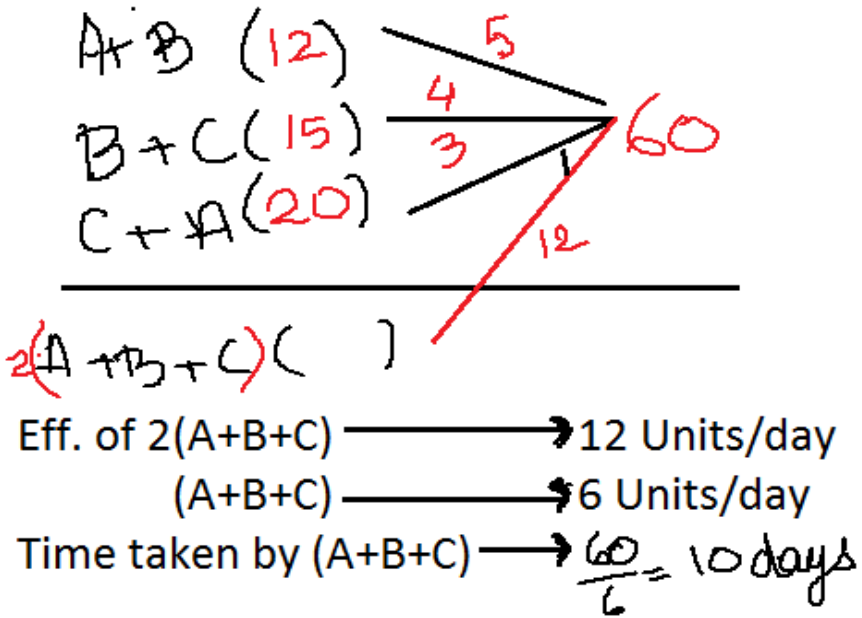


EXPALNATION

A and B can do a work in 12 days, B and C can do the same work in 15 days. C and A in 20 days. If A, B and C work together, they will complete the work in:

a) 5 days
b) 47 /6 days
c) 10 days
d) 47/3days

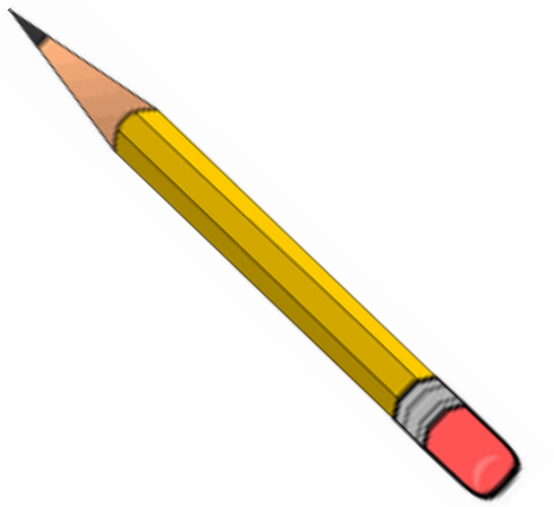
SOLUTION



PROBLEM STATEMENT 9

A and B can do a work in 8 days, B and C can do the same work in 12 days. A, B and C together can finish it in 6 days. A and C together will do it in:

a) 4 days
b) 6 days
c) 8 days
d) 12 days

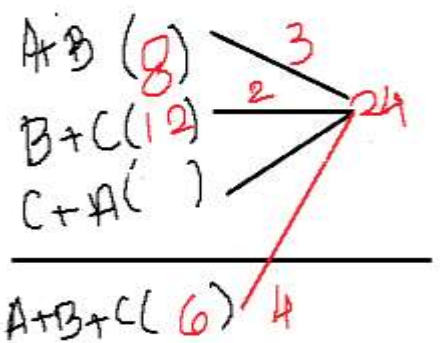


EXPLANATION

A and B can do a work in 8 days, B and C can do the same work in 12 days. A, B and C together can finish it in 6 days. A and C together will do it in:

a) 4 days
b) 6 days
c) 8 days
d) 12 days

SOLUTION



Eff. of (A+B+C) ⟶ 4

2(A+B+C) ⟶ 2 * 4 ⟶ 8units/day

Eff. of (A+c) = 2(A+B+C) - {(A+b)+(B+C)}

= 8 - (3+2) = 3 Units/day

A+C together can finish the work in ⟶ $\frac{24}{3}$ = 8days